DIRECTIONS *for questions 1 to 5:* Answer the questions on the basis of the information given below.

Gautam was playing with eight cards, each of which was labelled with a different digit from 1 to 8. He arranged the eight cards in a straight line, from left to right, in a certain order. The eight-digit number thus formed by the cards, from left to right, is labelled N. He then shuffled the eight cards and arranged them in a straight line again, from left to right. The eight-digit number then formed by the cards, from left to right, is labelled M.

For each of N and M, the first digit of the number refers to the digit at the extreme left, while the eighth digit refers to the digit at the extreme right.

It is known that

i. the first digit of M is at least seven times the first digit of N.

ii. the eighth digit of N is the same as the fifth digit of M.

iii. the seventh digit of N is twice the fourth digit of M, which, in turn, is one more than the third digit of N.

iv. the sixth digit of N and the second digit of M are both 4.

v. the eighth digit of N and the eighth digit of M differ by 6.

vi. the fifth digit of M and the second digit of N differ by 5.

vii. the third digit of M and the fifth digit of N are both 5.

viii. the seventh digit of M and the seventh digit of N are not the same.

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

What is the three-digit number formed by the sixth, seventh and eighth digits of N, taken in that order from left to right?

Q2. DIRECTIONS *for question 2:* Select the correct alternative from the given choices.

What is the difference between the second digit of N and the second digit of M?

- a) 1
- b) 0
- c) **2**
- d) 4

Q3. DIRECTIONS *for question 3:* Type in your answer in the input box provided below the question.

What is the sum of M and N?

Q4. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.
What is the sum of the fourth digit of N and the sixth digit of M?

- a) 10
- b) 11
- c) **13**
- d) 14

Q5. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.
The first digit of N is the same as

- a) the fourth digit of M.
- b) the fifth digit of M.
- c) **the sixth digit of M.**
- d) the seventh digit of M.

DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.

In a factory, which manufactures widgets, there are exactly six processes, P1 through P6, that each widget must pass through. Some of the six processes have constraints and all the constraints must be adhered to. The following table provides the time required for each process to be completed and the constraints that are present for each process:

Process	Time Required (in min)	Constraints
P1	28	Must start after P3 ends. Must start at most 5 minutes after P3 ends.
P2	32	Must start before P4 ends. Must start at most 15 minutes before P4 ends.
P3	19	Must end exactly 10 minutes before P2 ends.
P4	14	Must start exactly 20 minutes after P6 starts.
P5	27	Must end before P3 ends. Must end at most 25 minutes before P3 ends.
P6	33	Must start at least 15 minutes before P4 ends.

Assume that any process can begin only at the beginning of a minute and any process will end only at the beginning of a minute. For example, any process can begin at either 10:30:00 (in hh:mm:ss) or at 10:31:00 (in hh:mm:ss) but not at any other time in between and any process which starts at 10:30:00 (in hh:mm:ss) and takes 30 minutes to complete will end at 11:00:00 (in hh:mm:ss).

Q6. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

If a widget is manufactured in the minimum possible time, which of the following processes can be the third to end?

a) P5

b) P6

c) **P3**

d) P2

Q7. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.
If a widget is manufactured in the minimum possible time, and the first process to start commenced at 10:00 AM, at what time will P2 start?

- a) 10:20 AM
- b) 10:19 AM
- c) **10:42 AM**
- d) 10:17 AM

Q8. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.
What is the maximum time (in minutes) required for manufacturing each widget?

- a) 84
- b) 90
- c) **92**
- d) 88

Q9. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices. If a widget is manufactured in the maximum possible time, what is the fifth process to start?

- a) P5
- b) P2
- c) **P3**

Q10. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.
What is the minimum time required for manufacturing each widget?

- a) 68
- b) 70
- c) **72**
- d) 57

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

Bhargav conducted a survey in which he asked three groups of people – Americans, Brazilians and Chinese – about the sport(s) that they play among, Hockey and Football. Each person in any group need not necessarily play a sport and can play any number of sports. Each group comprised exactly 120 members and each person was part of exactly one group. The following information is known about their responses:

i. The number of Brazilians who do not play either of the two sports is the same as the number of Chinese who play both the sports.

ii. The number of Americans who play only Football is twice the number of Chinese who do not play either of the two sports.

iii. Across the three groups, 40 people do not play either of the two sports, while 60 people play both the sports.

iv. 95 Chinese play exactly one sport, while 70 Americans play exactly one sport.

v. The number of Americans who play Hockey is 65, which is the same as the number of Brazilians who play Football.

Q11. DIRECTIONS *for question 11:* Select the correct alternative from the given choices.
What is the number of Americans who do not play either of the two sports?

a) 15

b) 10

c) **25**

d) 35

Q12. **DIRECTIONS** *for question 12:* Type in your answer in the input box provided below the question.

How many Brazilians play only Hockey?

Q13. DIRECTIONS *for question 13:* Select the correct alternative from the given choices.
The number of Brazilians who play both the sports is the same as the number of

a) Chinese who do not play either of the two sports.

b) American who play both the sports.

c) **Chinese who play only Football.**

d) Americans who play only Hockey.

Q14. DIRECTIONS *for question 14:* Type in your answer in the input box provided below the question.

The maximum number of people in a single group who do not play either of the two sports is

Q15. DIRECTIONS *for question 15:* Select the correct alternative from the given choices.
The number of Chinese who play only Hockey cannot be

- a) twice the number of Americans who play only Football.
- b) twice the number of Brazilians who play only Football.
- c) **twice the number of Americans who play only Hockey.**
- d) twice the number of Brazilians who play only Hockey

Q16. **DIRECTIONS** *for question 16:* Type in your answer in the input box provided below the questi

Q17. DIRECTIONS *for question 17:* Select the correct alternative from the given choices.
Which train has the least number of non-AC compartments?

- a) Deodhar
- b) Grand Trunk
- c) **Duronto**
- d) Cannot be determined

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

What is the difference between the number of AC compartments in Duronto and the number of non-AC compartments in Grand Trunk?

Q19. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

If each AC compartment can seat 25 persons, while each non-AC compartment can seat 90 persons, what is the minimum seating capacity (i.e., the number of persons that can be seated in all the compartments of a train) of any train?

- a) 710
- b) 740
- c) **770**
- d) 800

Q20. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.
In how many trains is the number of AC compartments more than 5?

- a) 1
- b) 2
- c) **3**
- d) More than 3